Causal Bayesian Network Construction for Patients on ECMO using Large Language Models

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What?

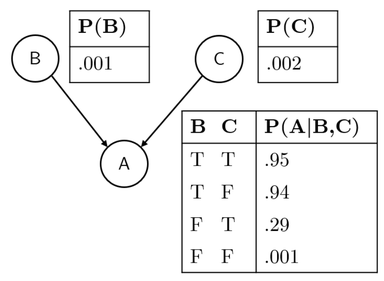
Problem Statement

Extracorporeal Membrane Oxygenation (ECMO) is a method for supporting patients with severe cardiac or respiratory failure. However, ECMO patients are at a higher risk of neurological injury. Understanding underlying causal mechanisms is critical for clinical decision-making. To this effect, we ask:

> What are the underlying causal relationships between the risk factors of neurological injury for patients on ECMO?

Background

Bayesian Networks (BN) factorize the joint distribution using a Directed Acyclic Graph (DAG).

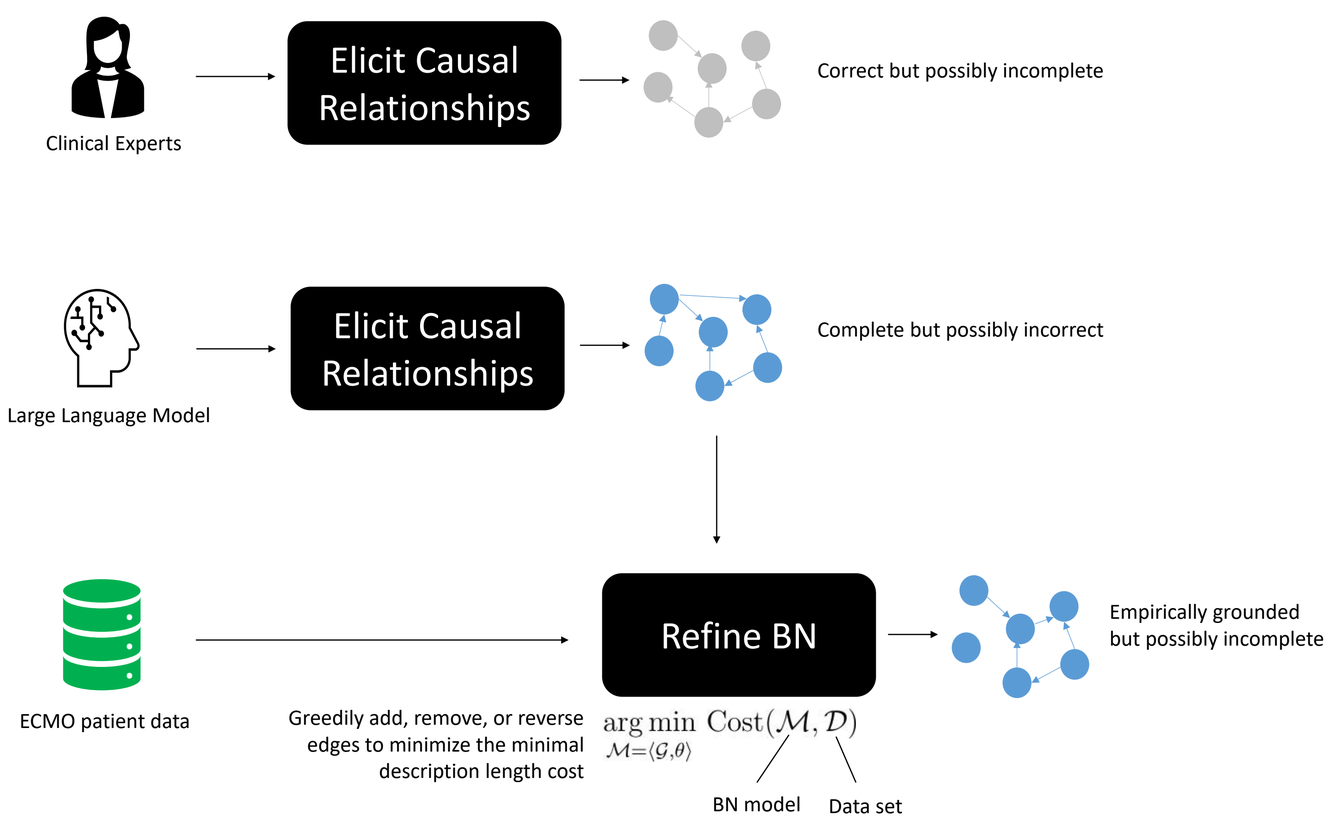


- Each edge $B \rightarrow A$ represents "B influences belief about A"
- Edges can be causal relations.
- Causality cannot be inferred purely from observational data..

Large Language Models capture knowledge, including causal relations, from their training data but they are stochastic and not amenable to reasoning.

How?

Methodology



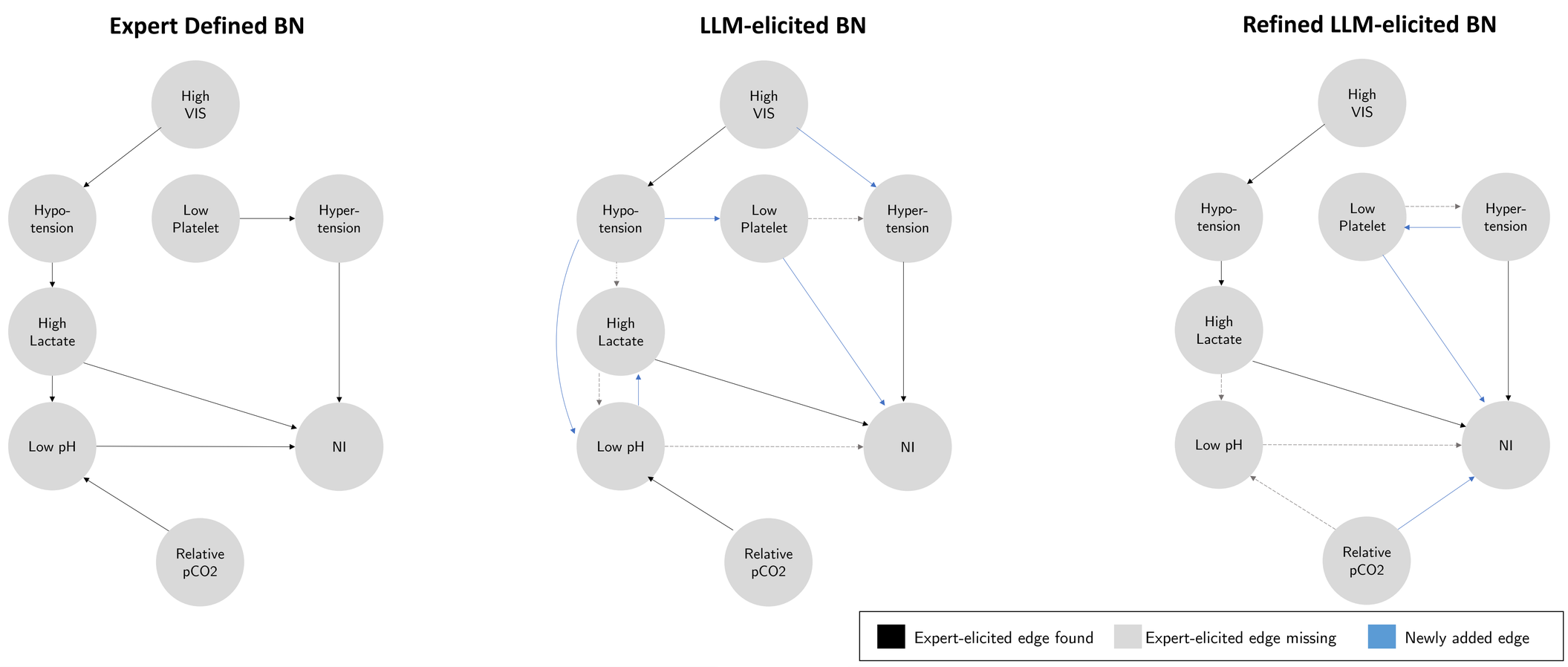
Data Description

Variable	Description	Frequency
High VIS	Vasoactive-inotropic score > 10	21.13%
Hypotension	Mean Arterial Pressure < 10th percentile for age	42.25%
Hypertension	Mean Arterial Pressure > 90th percentile for age	36.61%
Low Platelet	Platelet count < 50k/mm^3	33.8%
High Lactate	Lactate > 3 mmol/Liter	61.97%
Low pH	pH < 7.15	9.86%
Relative pCO2	Relative change in pCO2 from pre-ECMO to 24 hours on ECMO > 30%	40.84%
NI	Neurological Injury	25.35%

Data source: Children's medical center, Dallas.
Scope: 71 patients on ECMO from 0 to 24 hours.
Note: all variables except Relative pCO2 and SNI are considered true if the corresponding condition holds at any point.

Results

Empirical evaluation



Future work

- Testing causality of edges added during refinement.
- Scaling to larger variable spaces via relational models.
- Extending analysis to explicitly model relationships across time, e.g., pre-ECMO to on-ECMO.
- Inferring qualitative knowledge, e.g., monotonic and synergistic influences.

Acknowledgements

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Koller and Friedman 2009; Friedman and Goldszmidt 1996; Sprites et al. 2000; Mathur et al. 2024; Zečević et al. 2023; Shah et al. 2023